From Body Responses to Stress Recognition: An Explainable Wearable ML Framework

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Abstract

Can psychological stress be reliably identified through physiological signals collected from wearable devices? This study explores the relationship between emotional states and bodily responses using the WESAD dataset, focusing on the identification of three conditions: baseline, stress and amusement. The project aims to provide a compact but interpretable framework for wearable stress detection, combining physiological computing, machine learning, and behavioural psychology. The analysis builds upon the experimental design and baseline findings of the original WESAD study [1].

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Introduction

Understanding psychological stress through objective physiological measurements has become a central topic in affective computing and behavioural psychology. Stress is a complex psychophysiological state involving coordinated responses of the autonomic nervous system, which manifest in measurable changes across cardiovascular, electrodermal, thermal, and motor systems [2]. The increasing availability of multimodal wearable sensing has enabled the collection of rich physiological datasets, allowing researchers to investigate emotional states in a controlled but realistic experimental setting.

In this context, this project uses the WESAD Dataset to study the discrimination between stress, amusement and the baseline condition. First, the validity of the experimental conditions is verified through the replication of self-report questionnaire analysis from the original study. Second, physiological signals are segmented into fixed-length windows and transformed into feature representations. Three different window sizes (10, 30, and 60 seconds) were evaluated during preliminary experiments; however, results remained consistent across settings. Therefore, all reported analyses refer to the 60-second window configuration.

Finally, a Random Forest classifier is trained and interpreted using Mean Decrease in Impurity (MDI) and Permutation Feature Importance (PFI) to identify the most relevant physiological predictors of stress.

1. Dataset Description

The analysis is based on the WESAD Dataset, a publicly available multimodal dataset designed for stress and affect recognition in wearable computing. The dataset was collected in a controlled laboratory setting using standardized affect induction protocols, alternated with meditation sessions, alongside continuous physiological monitoring via wearable sensors. In fact, it includes synchronized recordings from two wearable devices: a chest-worn unit and a wrist-worn device. Multiple physiological signals are provided, including electrocardiography (ECG), respiration (RESP), electrodermal activity (EDA), blood volume pulse (BVP), skin temperature (TEMP), and accelerometer data (ACC).

In addition to physiological recordings, the dataset provides self-report questionnaire data, including PANAS, STAI, SAM (valence and arousal), and SSSQ, which are used to validate the effectiveness of the affect induction procedure and the perceived emotional states of participants. For more details see [3].

Overall, WESAD constitutes a well-established benchmark for studying physiological correlates of stress and developing machine learning models for affect recognition from multimodal biosignals.

2. Self-report Validation of Affective Conditions

Unlike the original study, where questionnaire-based validation is reported only briefly, this work provides a more detailed analysis of the self-report data to more accurately assess the psychological validity of the induced affective states.

First, questionnaire scores were computed for each participant by aggregating the corresponding item responses for each instrument. Subsequently, exploratory visualizations were generated to assess and validate the effectiveness of the experimental manipulation across subjects and affective conditions.

Figure 1 reports the distribution of self-reported valence and arousal scores across participants for each experimental condition.

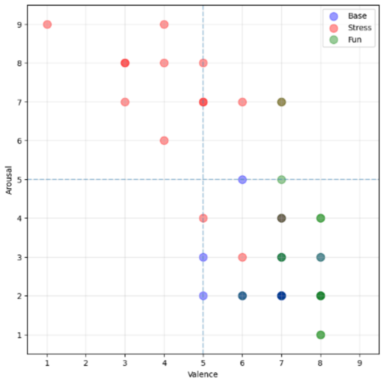

Figure 1 – Valence and Arousal scores across participants and conditions

The visualization highlights a clear separation between conditions, with the stress phase associated with lower valence and higher arousal values, while the amusement condition shows higher valence and moderate arousal.

Figure 2 shows the distribution of Positive Affect (PA), Negative Affect (NA), and STAI scores across participants, alongside condition-wise means.

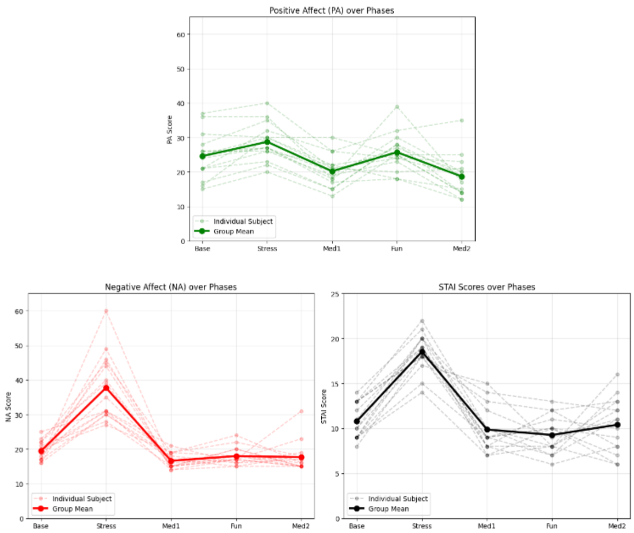

Figure 2 – Self-reported affect (PA, NA) and stress (STAI) across conditions

Positive Affect remains relatively stable across conditions, showing only a slight increase during both the stress and amusement phases compared with baseline and meditation. In contrast, Negative Affect and STAI scores are clearly higher during the stress condition, indicating a clear increase in perceived stress levels across participants.

The Subjective Stress Scale Questionnaire (SSSQ) was administered exclusively during the stress condition to characterize the qualitative nature of the induced stress. The results are summarized in Table 1.

	Engagement	Worry	Anger
Mean	3.9	3.7	1.6
Std	0.81	0.48	0.46

Table 1 – SSSQ dimensions' statistics (during the stress condition)

The results indicate that the induced stress state is mainly characterized by engagement and worry, while anger is substantially lower.

In conclusion, the results from the questionnaires confirm a clear separation between affective conditions. Participants consistently reported higher stress during the stress phase and more positive affect during the amusement phase.

This validation step ensures that the physiological signals used in the subsequent machine learning analysis are grounded in reliably induced and subjectively experienced emotional states, strengthening the overall interpretability and psychological relevance of the study.

3. Feature Extraction and Data Exploration

Raw physiological signals were processed through a window-based feature extraction pipeline to transform heterogeneous time-series data into a structured representation suitable for Machine Learning. For each signal modality, statistical descriptors were computed within fixed-length temporal windows, without applying global downsampling, to preserve the native dynamics of each signal. Specifically, for each signal segment the following features were extracted: mean (AVG), standard deviation (STD), median (MED), and slope coefficient (BET), capturing different aspects of central tendency, variability, and temporal trend of the signal.

Following feature extraction, an exploratory data analysis was conducted on the resulting feature space. First, the distribution of samples across conditions was examined, revealing an imbalance between classes, with a higher proportion of baseline segments compared to stress and amusement conditions.

Subsequently, feature distributions were analysed to assess their discriminative potential between classes. As an example of feature-level exploration, Figure 3 illustrates the distribution of wrist accelerometer magnitude (average on time window) across affective conditions.

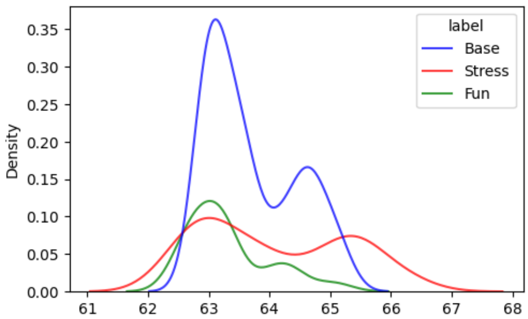

Figure 3 – Distributions of wrist accelerometer average across conditions

The visualization highlights lower values during the amusement condition compared to stress, suggesting distinct patterns of motor activity associated with different emotional states.

In addition, a correlation analysis between extracted features revealed a strong redundancy between mean (AVG) and median (MED) descriptors across many signal modalities. Therefore, most median-based features were excluded from the subsequent modelling phase to reduce feature redundancy and improve model interpretability.

4. Machine Learning Classification

To classify affective conditions from physiological features, a Random Forest classifier [4] from the Scikit-learn Python library [5] was adopted due to its robustness, ability to handle heterogeneous features, and intrinsic interpretability. Model optimization was performed through Grid Search cross-validation, exploring different combinations of hyperparameters.

After hyperparameter tuning, a validation curve was generated (Figure 4) to analyse the relationship between training and validation accuracy as a function of the number of estimators, allowing the assessment of model stability and generalisation capabilities.

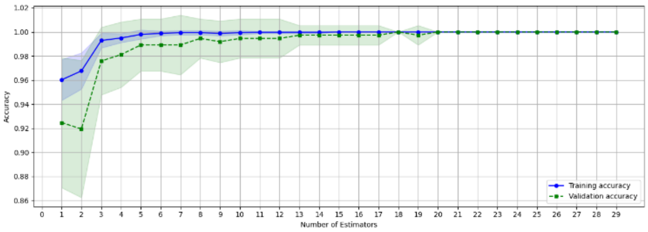

Figure 4 – Validation curve (accuracy vs number of estimators)

Model evaluation was conducted using accuracy, precision, recall, and F1-score with macro-averaging, together with confusion matrices for visual inspection of prediction performance (Figure 5).

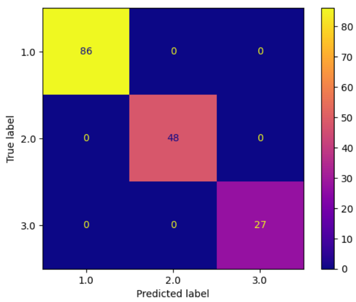

Figure 5 – Confusion matrix

The model achieved perfect classification results across all evaluation metrics (1.00), suggesting that physiological features extracted from multimodal biosignals contain highly discriminative information for distinguishing affective conditions. Unlike the original WESAD study, the proposed approach achieved substantially higher performance. This difference may be partially explained by differences in dataset construction and feature engineering: the original study employed a larger number of data points and a more complex processing pipeline, whereas the present work relies on a more compact feature representation based on aggregated statistical descriptors extracted from temporal windows.

5. Explainable AI Analysis

To improve the interpretability of the classification process, Explainable Machine Learning techniques were applied to the optimized Random Forest model. Feature importance was analysed using Mean Decrease in Impurity (MDI) [4] and Permutation Feature Importance (PFI) [6], allowing the identification of the physiological descriptors that contributed most to the discrimination between affective conditions.

First, MDI scores were extracted directly from the Random Forest model to evaluate how strongly each feature contributed to reducing classification uncertainty during the tree construction process. The resulting feature importance distribution is reported in Figure 6.

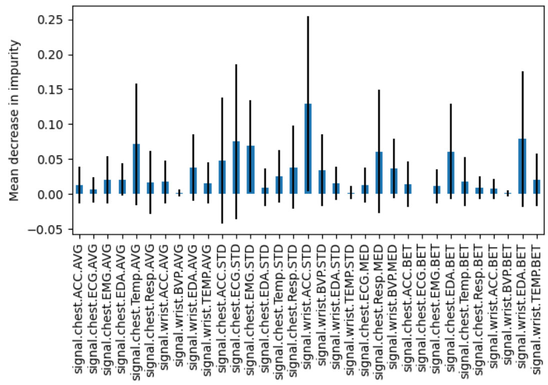

Figure 6 – MDI for each feature

The plot highlights that the most informative feature is the wrist accelerometer standard deviation, indicating that variability in motor activity plays a key role in distinguishing affective states. The remaining features show a more distributed importance, with several physiological descriptors from different modalities contributing in a relatively balanced way.

Complementarily, Permutation Feature Importance was computed by measuring the decrease in model performance after randomly shuffling individual features. This approach provides a model-agnostic estimate of feature relevance based on predictive degradation rather than tree structure alone. The corresponding results are shown in Figure 7.

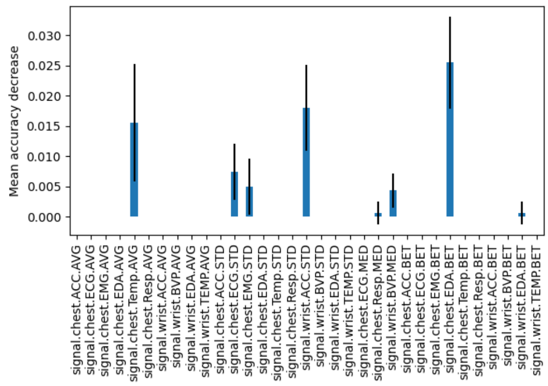

Figure 7 – PFI for each feature

In this case, the most influential features are chest EDA slope coefficient, followed by wrist acceleration standard deviation and chest temperature average, indicating that electrodermal activity and thermal responses play a central role in classification performance, together with motor activity variability.

Overall, both methods highlighted a subset of highly informative physiological descriptors, providing a more interpretable understanding of how different biosignals contribute to stress recognition.

Conclusions

This project investigated whether psychological stress can be identified from physiological signals using the WESAD Dataset, integrating signal processing, Machine Learning, and explainable AI.

Self-report questionnaires confirmed a clear separation between stress and amusement conditions, supporting the validity of the induced affective states. Physiological signals were then transformed into window-based features and used to train a Random Forest classifier optimized via grid search and cross-validation.

The model achieved very high classification performance across all settings, indicating strong discriminative power of the extracted features. Explainable AI analysis (MDI and PFI) further highlighted the most relevant physiological signals, particularly those related to electrodermal activity, body temperature, and motor variability.

Overall, the results show that stress and amusement can be effectively distinguished from multimodal physiological data, while interpretability methods provide additional insight into the most relevant physiological responses associated with stress.

Reproducibility

To facilitate reproducibility, all code used for data preprocessing, feature extraction, model training, and explainability analysis is available in a public repository [7]. The experiments were conducted in Python using Google Colaboratory, relying exclusively on the free computational resources provided by the platform and fixed random seeds to ensure consistent results across executions.

References

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